Humans Make AI Better

Podcast

Summary

Approximately Correct host, Scott Lilwall sits down with Amii Fellow and Canada CIFAR AI Chair, Matthew Taylor to discuss the "Human-in-the-Loop" (HITL) framework. Taylor, a reinforcement learning researcher describes his path into AI as driven largely by curiosity and fun. He began working in reinforcement learning in 2003 because it allowed systems to discover solutions on their own, often in surprising ways. What fascinated him most was watching "emergent" solutions, strategies no human explicitly programmed, developed through trial and error. Over time, reinforcement learning shifted from being purely academic video game research to having real-world impact, which Taylor finds especially exciting.

The conversation dives into the central theme of "human-in-the-loop", which Taylor defines as a collaborative approach where humans and AI both leverage their respective strengths to achieve a better outcome than either could alone. Humans excel at perception, common sense, and contextual understanding, while machines are strong at computation and optimization. Rather than trying to replace people, these collaborative frameworks aim to combine these strengths. When asked which application human-in-the-loop would be the most useful, Taylor explains that there are certain areas where human-in-the-loop is especially important, like medicine, where people want a human expert to supervise AI decisions to ensure safety, trust, and accountability.

Taylor strongly believes that humans should be involved from the very beginning of system design, not just at the end to "rubber-stamp" decisions. He gives the example of working with flight instructors on a pilot-training system. Without their input, he would have designed training in a way that ignored how pilots are actually taught, such as starting with basic skills like flying "straight and level". Collaborating with subject-matter experts early and continuously helps shape a much better system and shows how human input can fundamentally change the direction of a project.

He also explains why reinforcement learning, in particular, really needs humans in the loop. Agents are very good at optimizing whatever goal we give them, but humans are the ones who decide what that goal is. When goals are poorly defined, systems can behave in ways that technically "succeed" but completely miss the point. These moments expose the gap between what we think we said and what we actually told the system to do. Humans have to stay involved, constantly shaping and reshaping goals with ethical judgement, real-world experience, and common sense, things machines don't naturally have. Importantly, this isn't a one-time set-up. It's an ongoing process where we learn what we truly care about by watching how the system interprets our instructions.

Finally, Taylor discusses trust, explainability, and future directions. For humans to collaborate well with AI, they need to understand where systems are strong, where they are weak, and why they make certain decisions. Explanations should be tailored to what users actually need to know to make a decision or action. In high-stakes areas like healthcare, energy, and transportations, human-AI teamwork can lead to better results faster than full automation. Rather than pursuing full autonomy, Taylor argues that the most practical and powerful approach is thoughtful collaboration between human and AI systems.

Key Terms

Term	Definition
Reinforcement Learning	A type of AI system that improves through interaction, choosing actions, observing the outcomes, and adjusting its behavior to increase long-term success.
Agent	An AI agent is the brain of the system. Unlike a standard computer program that just follows a fixed set of instructions, an agent is designed to perceive its environment and take actions to reach a specific goal.
Reward Function	The formal way a system defines success, telling the AI what to optimize and strongly shaping how it behaves.
Human-in-the-Loop	A framework approach where humans remain involved at three key stages: defining goals, guiding learning, monitoring performance, and intervening when needed.
Emergent Behavior	New strategies or behaviors that arise from learning processes, even though they were not explicitly programmed by designers.
Explainable AI	Techniques that focus on making the internal logic and output of machine learning models transparent and interpretable to human users.
Markov Decision Process	A discrete-time mathematical framework used for modeling decision-making in situations where outcomes are partly random and partly under the control of a decision-maker.
Algorithmic Bias	Systematic and repeatable errors in a computer system that create unfair outcomes, typically stemming from biased training data, flawed model assumptions, or inappropriate optimization goals.

3 Key Takeaways

1. **AI is most effective as a teammate, not as a replacement.** The core of Human-in-the-Loop is recognizing the strengths and weaknesses of both humans and computer systems and utilizing these respective strengths to gain higher performance.

2. **Humans are essential for "problem defining", not just "problem solving"**. Machine Learning agents lack the moral and logical compass that humans possess. Humans involvement is necessary from the start to ensure AI's objectives align with real-world requirements and safety.

3. **Collaborative trust requires explainability.** Trust is built through iterative collaboration and for Human-in-the-Loop to work (especially in high-stakes fields like medicine or flight training), AI must be explainable. The user needs to be able to understand they why behind a decision so they can identify when the AI is operating inaccurately or outside of its expertise.

Discussion Questions

1. Matthew Taylor mentions that humans and AI have complementary strengths (e.g., computers excel at math and humans have common sense). In your specific industry, what are the "long division" tasks that could be handed off to AI, and what are the "vision/common sense" tasks that should be left to humans?

2. Taylor argues that a doctor shouldn't just "rubber stamp" an AI's decision at the end of a decision making process. How might we design our processes so that the person "in the loop" remains critically engaged?

3. One of the weaknesses of AI highlighted in the podcast is contextual/background knowledge. In Taylor's example, he explains that an AI system couldn't predict a power spike caused by a football match because it didn't "know" about the event. What are the external events or common-sense factors in your industry that AI would be blind to? How could you circumvent this?

Relevant Resources

Resources related to the topics explored in the "Human's Make AI Better" podcast:

1. "Reinforcement Learning Inherently Needs a Human" Paper

- Retzlaff, C. O., Dasgupta, S., Waytowich, N., & Taylor, M. E. (2024). *Reinforcement learning inherently needs a human: A survey and position on requirements, challenges, and opportunities*. arXiv. https://doi.org/10.48550/arXiv.2405.00746

2. "Collaborative Intelligence" Paper

- Taylor, M. E., & Scott, S. (2025). *Collaborative intelligence: The future of human-in-the-loop reinforcement learning*. arXiv. https://arxiv.org/abs/2504.17006 *(Note: Authors and titles for ArXiv preprints vary by version; please verify the specific author list if this is a shared draft.)*

3. Matthew Taylor YouTube Video

- Amii. (2023, July 12). *Human in the loop with Matthew E. Taylor | Amii* [Video]. YouTube. https://www.youtube.com/watch?v=04lH5odVC24

References

Alberta Machine Intelligence Institute (Amii). (2024, August 2). *Humans Make AI Better with Matt Taylor | Approximately Correct Podcast*. Amii. https://www.amii.ca/videos/approximately-correct-humans-make-ai-better

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